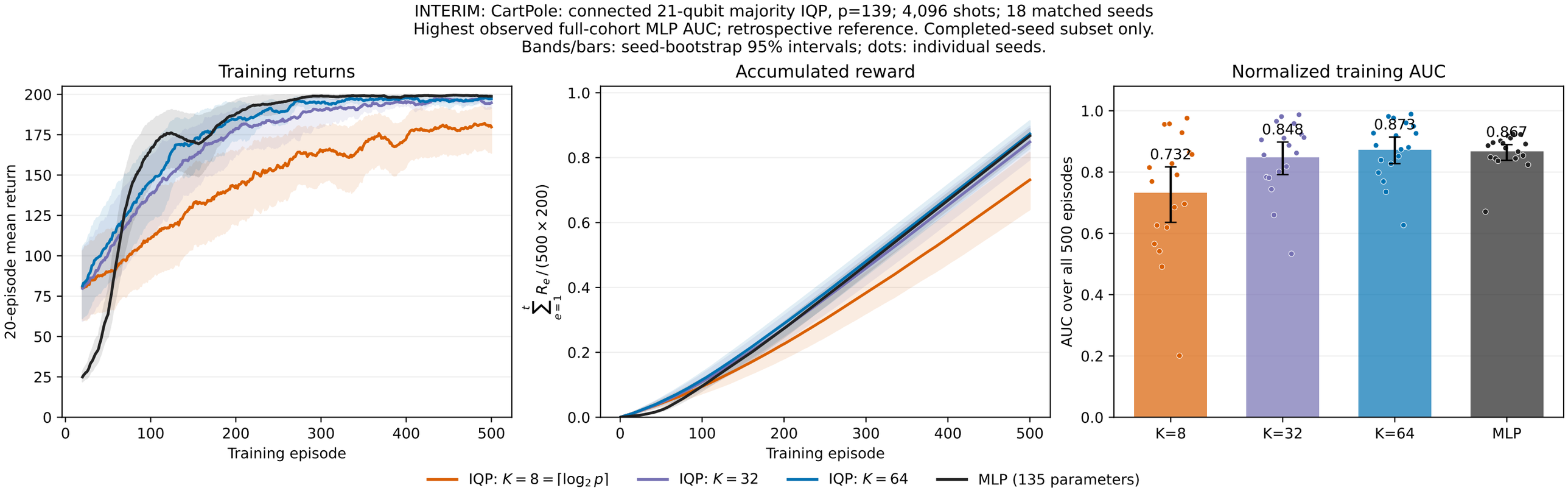

INTERIM: CartPole: connected 21-qubit majority IQP, p=139; 4,096 shots; 18 matched seeds
Highest observed full-cohort MLP AUC; retrospective reference. Completed-seed subset only.
Bands/bars: seed-bootstrap 95% intervals; dots: individual seeds.
Training returns
20-episode mean return
Training episode
Accumulated reward
$\sum_{e=1}^{t} R_e / (500 \times 200)$
Training episode
Normalized training AUC
AUC over all 500 episodes
0.732
0.848
0.873
0.867
K=8
K=32
K=64
MLP
IQP: $K = 8 = \lceil \log_2 p \rceil$
IQP: $K = 32$
IQP: $K = 64$
MLP (135 parameters)